

1.What is a variable in JavaScript?

A variable is a container used to store data or values in JavaScript

2.What are the three keywords used to create variables?

let,var,const

3.Write the syntax to create a variable using var.

```
var variableName = variable; var a = 10;
```

4.Write the syntax to create a variable using let.

```
let variableName = variable; let a = 20;
```

5.Write the syntax to create a variable using const.

```
const variableName = variable; const a = 40;
```

6.What is declaration?

declaration is a process of creating a variable without assigning a values to it.

```
var a;
```

7.What is initialization?

initialization is a process of assigning a value to a variable for the first time.

```
var a = 12;
```

8.What is reassignment?

reassignment is a process of assigning a new value to an already declared variable.

```
let age = 20;
```

```
age = 20;
```

9.What is redeclaration?

redeclaration is a process of declaraing a same variable again using the same keyword within the same scope

```
var a = 20;
```

```
var a = 30;
```

10.Which keyword allows redeclaration?

```
var
```

11.Which keyword allows reassignment?

```
let and let
```

12.Which keyword requires initialization when declared?

```
const
```

13.Identify the declaration and initialization:

```
let age = 25;
```

declaration--> let age

initialization-->25

14.What is the value of a?

```
var a = 100;  
console.log(a);  
100
```

15.Change the value of this variable to 200:

```
let number = 100;  
    number = 200;
```

16.What will be the output?

```
var a = 10;  
console.log(a);  
10
```

17.What will be the output?

```
var a = 10;  
a = 20;  
console.log(a);  
20
```

18.What will be the output?

```
var a = 10;  
var a = 30;  
console.log(a);  
30
```

19.Write a var variable named name with the value "John".

```
var name = "John";
```

20.Create a var variable named price with the value 500.

```
var price = 500;
```

21.Reassign price from 500 to 1000.

```
var price = 500;  
price = 1000;
```

22.What will be the output?

```
var x = 50;  
x = 100;  
console.log(x);  
100;
```

23.Can a var variable be reassigned?

```
Yes, var can be reassigned  
eg,  
var a = 10;  
a=20;
```

24.Can a var variable be redeclared?

```
Yes,var can be redeclared
```

25. Write an example of var redeclaration.

```
var a = 10;  
a=20;  
console.log(a);
```

26. Create a let variable named age with the value 25.

```
let age = 25;
```

27. What will be the output?

```
let age = 20;  
age = 30;  
console.log(age);  
30
```

28. Can a let variable be reassigned?

Yes, let can be reassigned

29. Can a let variable be redeclared?

No. let can be redeclared

30. Find the error:

```
let name = "John";  
let name = "David";  
  redeclaration of the name.
```

31. Create a let variable called city and assign "Chennai".

```
let city = "Chennai";
```

32. Change the value of city to "Salem".

```
let city = "Chennai";  
city = "Salem";
```

33. What will be the output?

```
let x = 10;  
x = 50;  
console.log(x);  
50
```

34. Write a let variable called salary with the value 25000.

```
let salary = 25000
```

35. Reassign salary to 30000.

```
let salary = 25000  
salary = 30000;
```

36. Create a const variable called pi with the value 3.14.

```
const pi = 3.14
```

37. Can a const variable be reassigned?

No, const cannot be reassigned

38. Can a const variable be redeclared?

No, const cannot be redeclared

39. What is wrong with this code?

```
const age;
```

```
age = 25;
```

A const variable must be initialized when it is declared

40. What happens here?

```
const price = 500;
```

```
price = 1000;
```

It will give an error because a const variable cannot be reassigned.

41. Create a const variable called country with the value "India".

```
const country = "India";
```

42. What will be the output?

```
const x = 100;
```

```
console.log(x);
```

100

43. Which keyword should you use if the value should not be reassigned?

const

44. What is the difference between let and const?

let → The value can be reassigned.

const → The value cannot be reassigned.

Both let and const cannot be redeclared in the same scope.

45. What is the difference between var and const?

var → Can be reassigned and redeclared.

const → Cannot be reassigned or redeclared.

var has function scope, while const has block scope.

46. Write JavaScript code to print Hello World using console.log().

```
console.log("Hello World");
```

47. Write JavaScript code to print the number 500 using console.log().

```
console.log(500);
```

48. What is the purpose of console.warn()?

console.warn() is used to display warning message in the browser console

49. What is the purpose of console.error()?

console.error() is used to display error message in the browser console

50. What is the purpose of each?

alert() -- display a popup message to a user

```
prompt() -- Displays a popup and takes input from the user.  
confirm() -- Displays a popup with OK and Cancel options.  
document.writeln() -- Prints content directly on the webpage.  
console.log() -- Prints output in the browser console.
```